Linear Separability

Why It Matters for Classification

CS 115B: Fundamentals of NLP

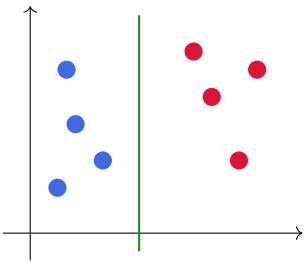
Brandeis University

Spring 2026

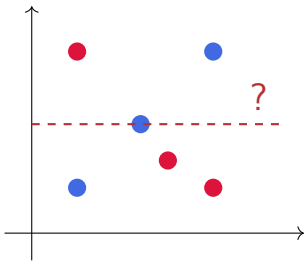
The Big Question

Given two classes of points...

Can you draw a straight line (or hyperplane) to separate them?



✓Yes!



× No!

Formal Definition

Linear Separability

A dataset $\{(\mathbf{x}_i, y_i)\}_{i=1}^{n}$ with $y_i \in \{-1, +1\}$ is **linearly separable** if there exists a weight vector $\mathbf{w}$ and bias b such that:

$$y_i \cdot (\mathbf{w}^\top \mathbf{x}_i + b) > 0 \quad \text{for all } i$$

In plain English:

- There's a hyperplane $\mathbf{w}^\top \mathbf{x} + b = 0$
- All $+1$ points are on one side
- All -1 points are on the other side
- No point lies exactly on the boundary

Dimensionality:

- In 2D: separating *line*
- In 3D: separating *plane*
- In d-D: separating *hyperplane* ($(d-1)$-dimensional)

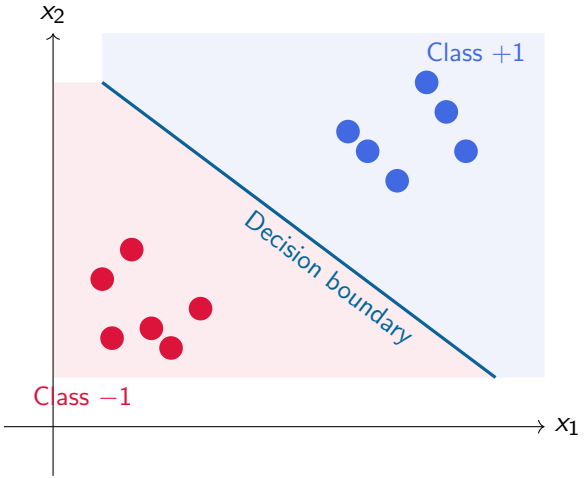
Why Does It Matter?

Linear classifiers (perceptron, logistic regression, linear SVM) can *only* learn linear decision boundaries.

If data is linearly separable	**If data is NOT separable**
Perceptron **will converge**	Perceptron **will never converge**
Can achieve 0% training error	Must accept some training errors
Linear classifier is appropriate	Need different approach

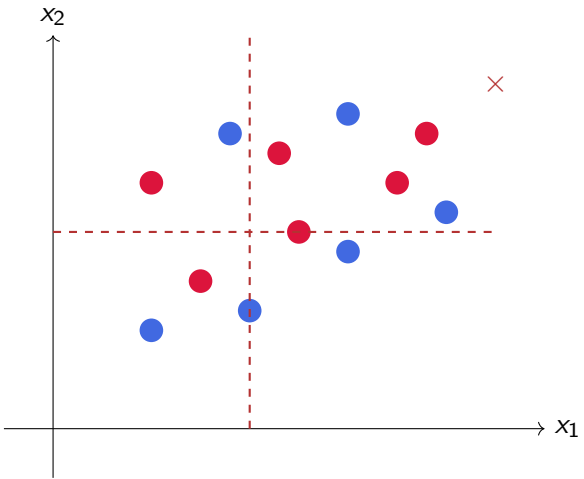
Linear separability determines whether a linear classifier can perfectly solve your problem.

Example: Linearly Separable Data



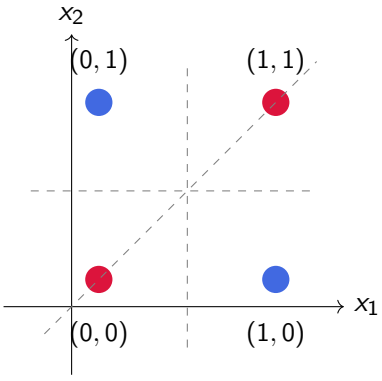
Linearly separable! A line cleanly divides the classes.

Example: NOT Linearly Separable (Interleaved)



NOT linearly separable! Classes are interleaved — no line works.

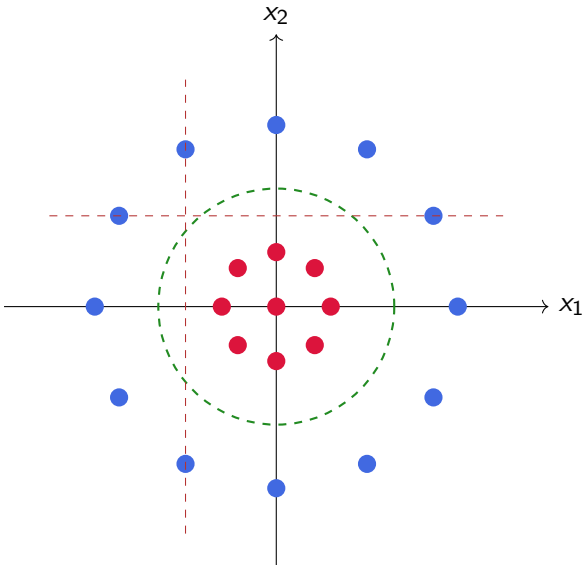
Example: NOT Linearly Separable (XOR Pattern)



x_1	x_2	XOR	Class
0	0	0	−
0	1	1	+
1	0	1	+
1	1	0	−

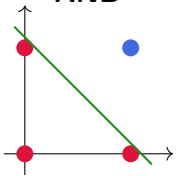
The classic example! XOR requires a *non-linear* boundary.

Example: NOT Linearly Separable (Circular)



Boolean Functions: Which Are Separable?

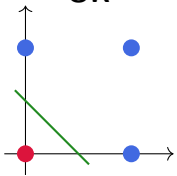
AND



✓Separable

$w_1 = 1, w_2 = 1, b = -1.5$

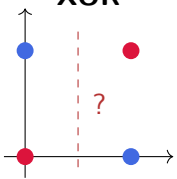
OR



✓Separable

$w_1 = 1, w_2 = 1, b = -0.5$

XOR



× NOT Separable

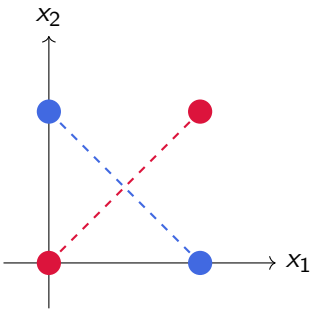
No solution exists!

Function	AND	OR	NAND	NOR
Separable?	✓	✓	✓	✓

Function	XOR	XNOR
Separable?	×	×

Why Can't a Line Separate XOR?

Intuition: The + points are "diagonally opposite"



Any line must:

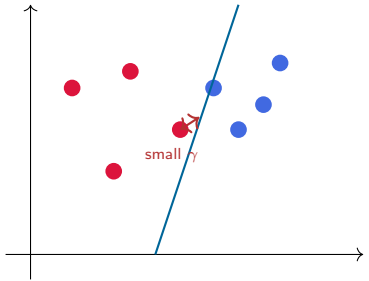
- Put $(0, 1)$ and $(1, 0)$ together
- Put $(0, 0)$ and $(1, 1)$ together

But the diagonals **cross**!

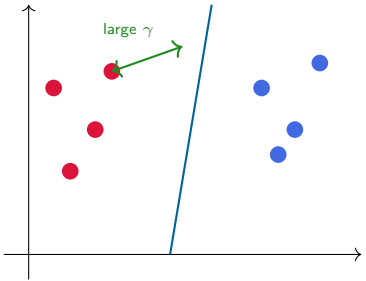
No line can separate crossing groups.

Formally: The convex hulls of the two classes intersect.

Not Just Separable — How *Well* Separable?



Small margin



Large margin

The margin γ = distance from nearest point to boundary

Larger margin $\Rightarrow$ easier to learn, better generalization

Margin and the Perceptron

Perceptron Convergence Theorem:
If data is linearly separable with margin γ and max radius R:

$$\text{Number of mistakes} \leq \left(\frac{R}{\gamma}\right)^2$$

Implications:

- Larger $\gamma \Rightarrow$ fewer mistakes
- Larger $\gamma \Rightarrow$ faster convergence
- If $\gamma = 0$: bound is infinite!

Extreme cases:

- γ large: Easy problem, fast learning
- γ tiny: Hard problem, slow learning
- $\gamma = 0$: Not separable, never converges

Support Vector Machines explicitly maximize γ.

When Data Isn't Linearly Separable

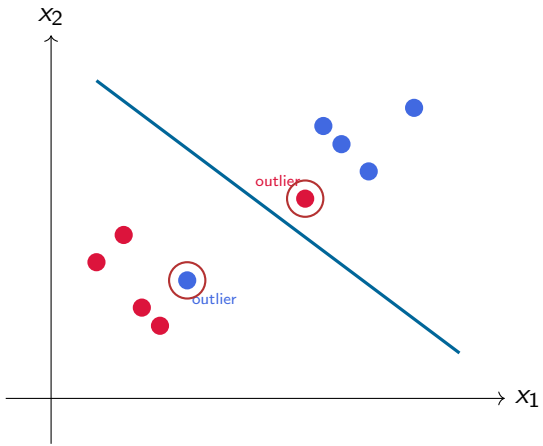
Three main strategies:

1. **Accept some errors** (soft margin)
   - Allow a few misclassified training points
   - Logistic regression, soft-margin SVM
   - Good when classes mostly separate with some overlap

2. **Transform the features** (feature engineering / kernels)
   - Add new features to make data separable
   - Example: Add $x_1 \cdot x_2$ to solve XOR
   - Kernel trick: implicit high-dimensional mapping

3. **Use a non-linear classifier** (neural networks)
   - Hidden layers learn non-linear boundaries
   - Can approximate any decision boundary
   - The deep learning approach

Solution 1: Accept Errors (Soft Margin)



Trade-off: Misclassify a few points to get a good overall boundary.
Used in: Logistic regression, soft-margin SVM

Solution 2: Feature Engineering

XOR becomes separable with the right features!

Original features: (x_1, x_2)

x_1	x_2	y
0	0	$-$
0	1	$+$
1	0	$+$
1	1	$-$

Not separable in 2D

Add feature: $x_3 = x_1 \cdot x_2$

x_1	x_2	x_1x_2	y
0	0	0	$-$
0	1	0	$+$
1	0	0	$+$
1	1	1	$-$

Separable in 3D!

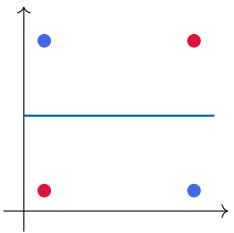
Solution: $w_1 = 1, w_2 = 1, w_3 = -2, b = -0.5$

$$\hat{y} = \text{sign}(x_1 + x_2 - 2x_1x_2 - 0.5)$$

Solution 3: Neural Networks

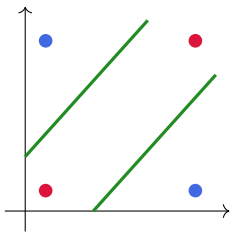
Hidden layers create non-linear boundaries

Single Layer



Can't solve XOR

Multiple Layers



Can solve XOR!

This is why we need deep learning!

Hidden layers learn useful features automatically.

Summary

Linear Separability:

- Data is linearly separable if a hyperplane can divide the classes
- Determines whether linear classifiers can achieve zero training error

Key Examples:

- AND, OR, NAND, NOR are linearly separable
- XOR is *not* linearly separable — the classic counterexample

The Margin:

- Measures *how well* separated the data is
- Larger margin $\Rightarrow$ easier learning, better generalization

When Data Isn't Separable:

- Accept errors (soft margin)
- Add features (feature engineering, kernels)
- Use non-linear models (neural networks)